

Variational Prefix Tuning for Diverse and Accurate Code Summarization

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```
public static String selectText (XPathExpression expr, Node context) {  
    try {  
        return (String)expr.evaluate(context, XPathConstants.STRING );  
    } catch (XPathExpressionException e) {  
        throw new XmlException(e);  
    }  
}
```

Transformer Model: evaluates the xpath expression as a text string.

Human Written: evaluates the xpath expression as a text.

Automatic Source Code Summarization

- Translate a sequence of source code tokens, represented as $\mathbf{x} = (x_1, \dots, x_n)$ into a sequence of natural language tokens $\mathbf{y} = (y_1, \dots, y_m)$ that aims to accurately reflect the core functionality and intent of the source code.
- Currently, Transformer-based Large Language Models for Code (LLMC) achieve SOTA performance in this field.
- Finetuning is a common approach to further improve LLMC's ability or align with user's generation preference.

Challenges



- Fine-tuning causes loss in generation diversity [Wang et al. 2022].
- Prior works only evaluate their method based on a single generation.
- Consider the case where the output is inadequate for the user and an alternative solution is needed.

What Options Do We Have?

- **Sampling:** Prevalent method for generating diverse outputs but came at a cost of accuracy.
- **Prompting:** LLM often gets too stubborn on its first thought and cannot self correct. [Huang et al. 2023, Liang et al. 2023]
- **Beam search:** Usually generate results that only varies by minor differences. [Kumar et al. 2004, Macherey et al, 2008]



How can we enhance the LLMC's ability to generate a set of diverse yet accurate outputs?

Variational Prefix Tuning

A novel approach to enhance LLM's ability in generating both
diverse and accurate output.

Motivation of Our approach

- Conditional Variational AutoEncoder (CVAE) model has been proven effective in other diverse generation tasks. [Wang et al. 2017, Jain et al. 2017]

Context	Examples of Generated Responses
thank your for calling world airline. what can I do for you? _eou_	Eg.1: i'd like to make a reservation for you, please Eg.2: do you know where i can get to get? Eg.3: can you tell me the way to the station?
how much is the rent? _eou_ the rent is \$1500 per month.	Eg.1: no problem. i'll take it. Eg.2: this one is \$1.50.50,000 yuan per month. Eg.3: that sounds like a good idea.
guess who i saw just now ? _eou_ who? _eou_ john smith. _eou_ that bad egg who took the low road since he was a boy.	Eg.1: it is my favorite. Eg.2: no, but i didn't think he was able to get married. i had no idea to get her. Eg.3: this is not, but it's not that bad. it's just a little bit, but it's not too bad.

Dialog
Generation



Predicted Object Labels:
'cup' 'fork' 'knife' 'sandwich' 'dining table' 'mouse'

AG-CVAE:
a close up of a plate of food on a table
a table with a plate of food on it
a plate of food with a sandwich on it
a white plate topped with a plate of food
a plate of food on a table next to a cup of coffee

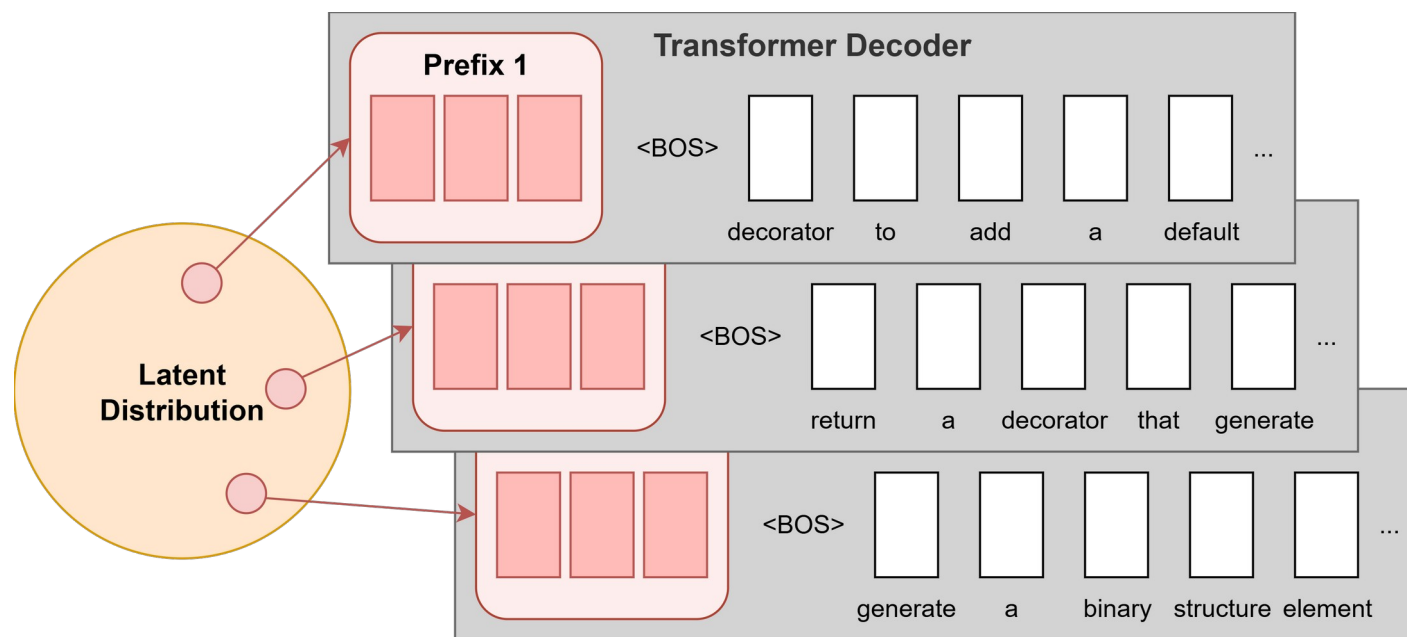
LSTM Baseline:
a close up of a plate of food on a table
a close up of a plate of food with a sandwich
a close up of a plate of food
a close up of a plate of food on a white plate
a close up of a plate of food with a sandwich on it

Image Caption

Can They Work With LLMs?

- Similar work (Transformer-CVAE models) exists. [Wang et al. 2019, Fang et al. 2021]
 - Requires a complete retraining of the entire model.
 - No access to the extensive knowledge learned by the LLMs.
- Training an LLM requires extensive amount of resource.
- Can we integrate CVAE as a modular component?
 - No need to optimize the LLM parameters when training this CVAE module.

Motivation of Our Approach



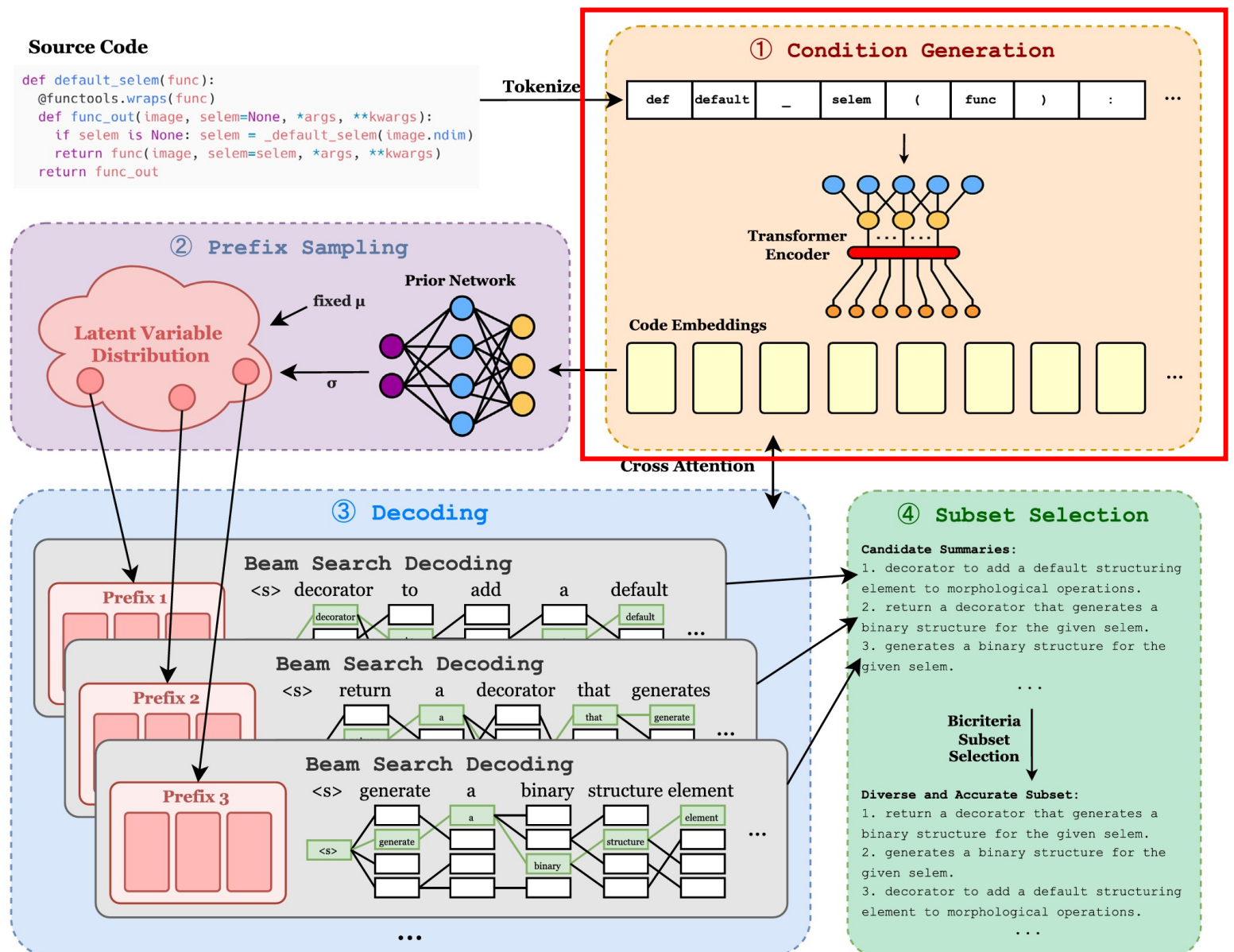
- Inspired by Prefix Tuning. [Li et al. 2021]
- Uses a variable set of vectors sampled from a learned distribution as Prefixes.
- Allows Integration of CVAE framework into existing pre-trained model in a parameter-efficient way.



Contribution of Our approach

1. Integrating CVAE framework into LLM in a modular way, eliminating the need to retrain the entire LLM.
2. A novel pipeline for enhancing LLM's ability to generate a set of diverse yet accurate summaries.

An Overview of Our Approach



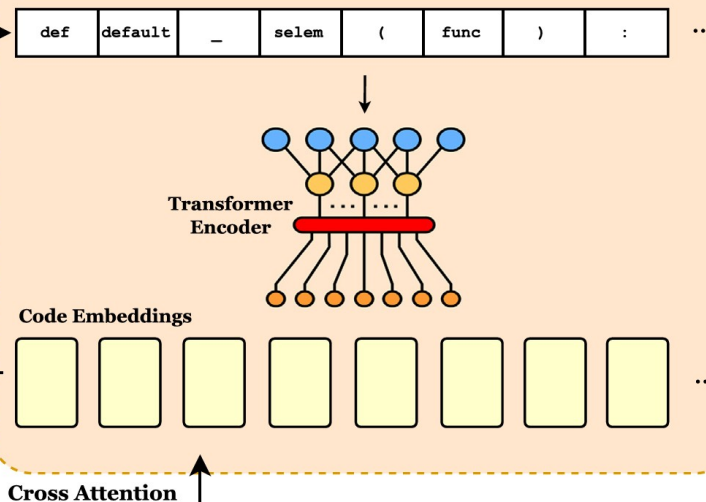
An Overview of Our Approach

Source Code

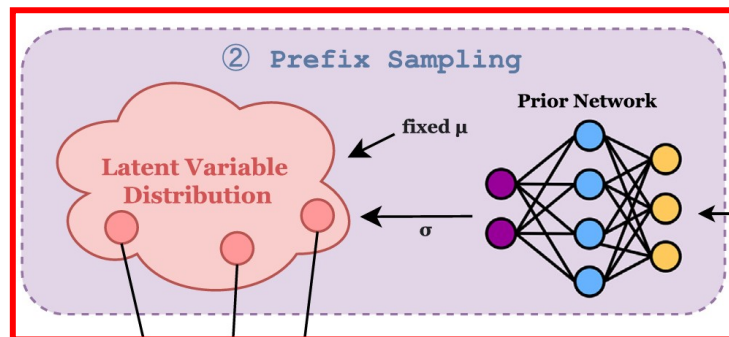
```
def default_selem(func):
    @functools.wraps(func)
    def func_out(image, selem=None, *args, **kwargs):
        if selem is None: selem = _default_selem(image.ndim)
        return func(image, selem=selem, *args, **kwargs)
    return func_out
```

Tokenize

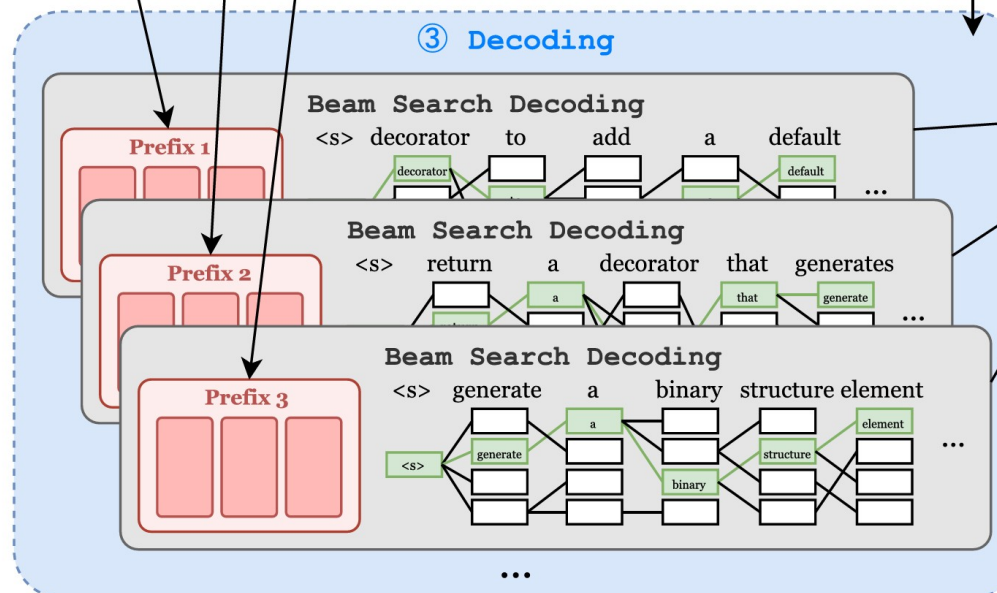
① Condition Generation



② Prefix Sampling



③ Decoding



④ Subset Selection

Candidate Summaries:

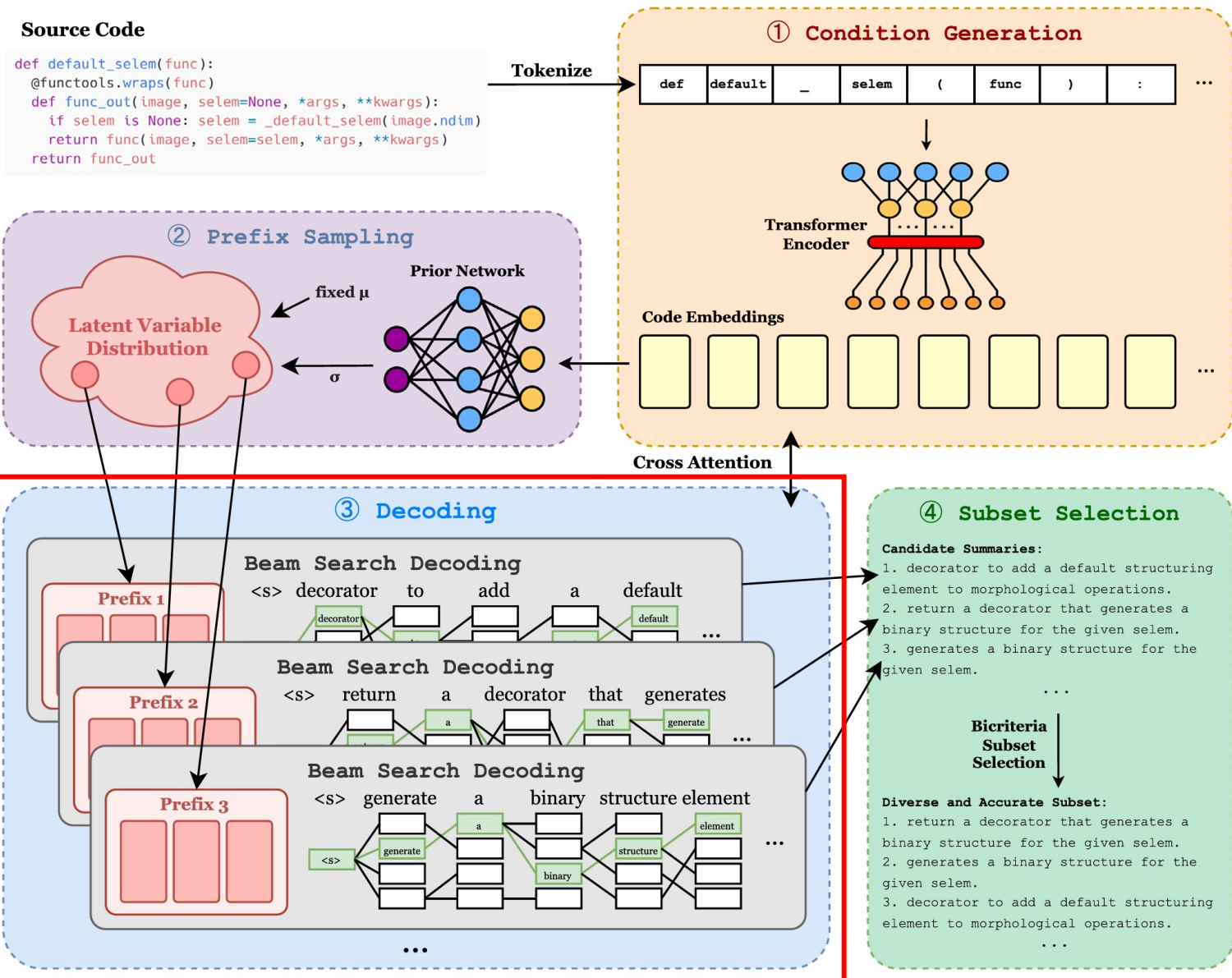
1. decorator to add a default structuring element to morphological operations.
2. return a decorator that generates a binary structure for the given selem.
3. generates a binary structure for the given selem.

Bicriteria Subset Selection

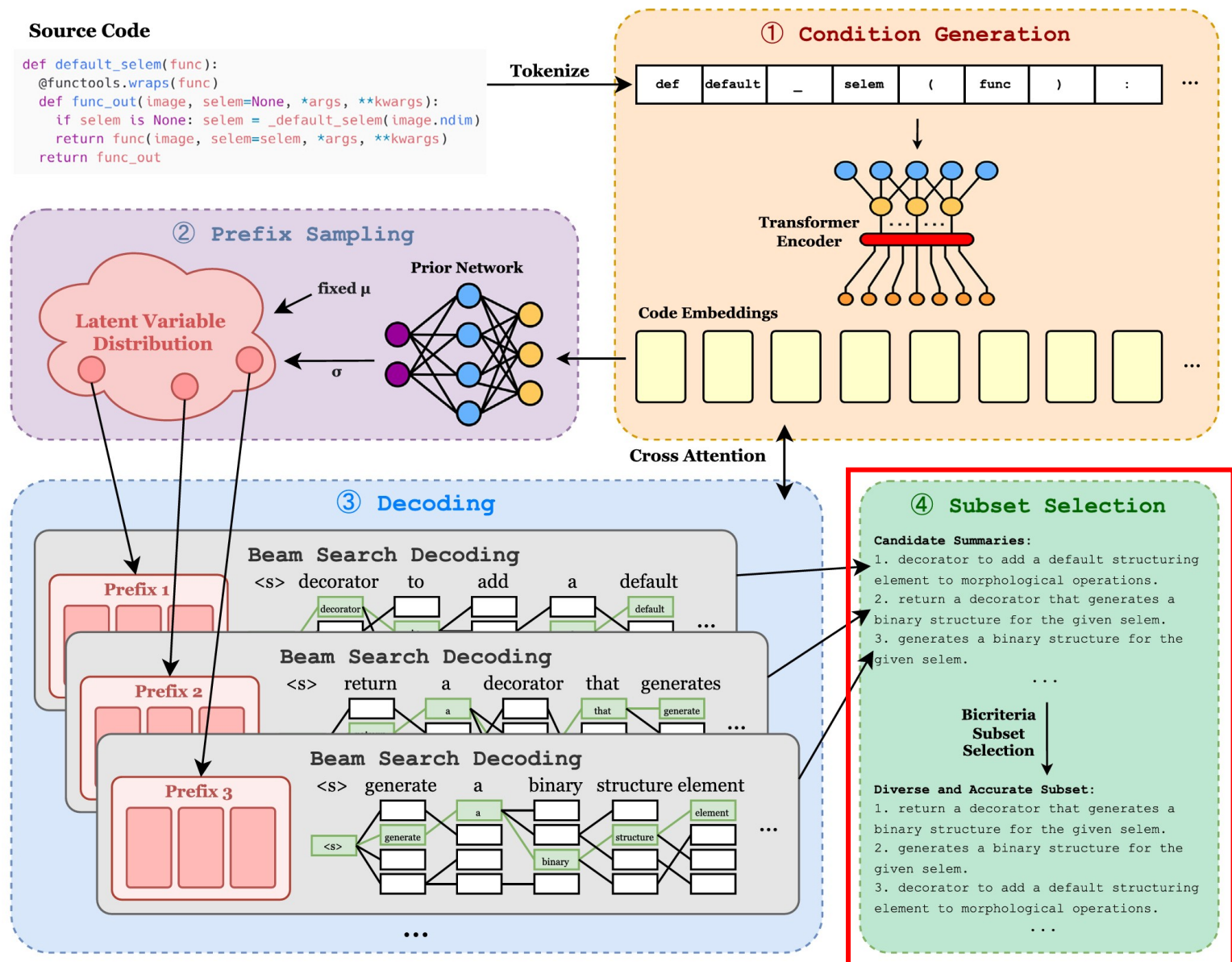
Diverse and Accurate Subset:

1. return a decorator that generates a binary structure for the given selem.
2. generates a binary structure for the given selem.
3. decorator to add a default structuring element to morphological operations.

An Overview of Our Approach



An Overview of Our Approach



Bi-criteria Subset Selection

A bi-criteria subset selection is adopted to further enhance the diversity and accuracy of the final set.

- Select a subset s from a larger set S that maximize objective, $\alpha g(s) + \beta h(s)$,
- $g(s)$ is the sequence score (normalized log probability) of the sequence, $h(s)$ is presented by one minus pair-wise bleu score.

④ Subset Selection

Candidate Summaries:

1. decorator to add a default structuring element to morphological operations.
2. return a decorator that generates a binary structure for the given selem.
3. generates a binary structure for the given selem.

...

**Bicriteria
Subset
Selection**



Diverse and Accurate Subset:

1. return a decorator that generates a binary structure for the given selem.
2. generates a binary structure for the given selem.
3. decorator to add a default structuring element to morphological operations.

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Evaluation

- Baselines:
 - Sampling
 - Beam Search
 - Diverse Beam Search (DBS) [Vijayakumar et al, 2018]
 - Stochastic Beam Search (SBS) [Kool et al, 2019]
- Backbone model
 - CodeT5+
- We use oracle metrics to evaluate the performance.
 - Consider diverse generation task as creating a list of options.
 - Maximize the likelihood of containing at least one adequate output.

Performance Compared To Baselines

- We apply VPT and other baselines to current State-Of-The-Art code summarization model CodeT5+
- We first generate a large set of 100 summaries then select subsets with size 10 and 20.
- Our Method shows superior performance across all metrics

Lang	Method	#U	B	R	M	BT-PR	BT-RE	BT-F1	SIDE
Python	Beam	10	44.28	61.50	<u>42.14</u>	93.63	93.00	93.14	93.28
	Sampling		<u>44.31</u>	<u>62.35</u>	41.50	<u>93.88</u>	<u>93.19</u>	<u>93.26</u>	<u>94.27</u>
	SBS		42.30	59.76	40.50	93.25	92.76	92.80	93.33
	DBS		43.07	59.90	40.53	93.28	92.58	92.81	92.54
	VPT		46.40	63.83	43.52	94.16	93.38	93.62	95.21
	Beam	20	45.63	63.12	<u>43.57</u>	93.85	93.33	93.40	94.53
	Sampling		<u>45.84</u>	<u>64.08</u>	42.95	<u>94.27</u>	<u>93.50</u>	<u>93.58</u>	<u>95.16</u>
	SBS		43.10	60.84	41.56	93.40	93.01	93.00	94.43
	DBS		44.49	61.81	42.76	93.64	93.14	93.17	94.56
	VPT		48.62	66.30	45.34	94.75	93.76	93.96	96.27
	Beam	100	47.13	64.70	45.37	94.01	93.71	93.59	96.10
	Sampling		48.92	67.25	45.77	94.97	93.95	94.15	95.97
	VPT		51.86	69.83	47.75	95.78	94.23	94.60	97.18
Java	Beam	10	<u>47.67</u>	62.28	<u>45.12</u>	93.80	<u>93.21</u>	<u>93.36</u>	91.59
	Sampling		<u>47.46</u>	<u>62.41</u>	44.63	<u>93.93</u>	93.12	93.34	<u>92.42</u>
	SBS		46.65	61.30	44.48	93.57	93.04	93.14	91.55
	DBS		47.11	61.54	44.73	93.66	93.11	93.21	91.72
	VPT		49.22	64.46	45.83	94.52	93.40	93.72	93.55
	Beam	20	<u>48.11</u>	63.07	<u>45.81</u>	93.90	<u>93.42</u>	<u>93.49</u>	92.72
	Sampling		<u>48.10</u>	<u>63.24</u>	45.22	<u>94.10</u>	93.28	<u>93.49</u>	<u>93.09</u>
	SBS		46.93	61.73	46.93	93.62	93.20	93.20	92.46
	DBS		47.93	62.59	45.70	93.83	93.34	93.39	92.99
	VPT		50.79	66.12	47.30	94.86	93.78	94.05	95.02
	Beam	100	48.46	63.42	46.51	93.84	93.55	93.46	93.56
	Sampling		48.68	64.06	45.79	94.28	93.43	93.65	94.40
	VPT		52.60	68.01	48.85	95.20	94.11	94.33	96.47

Qualitative Examples

Source code:

```
def default_selem(func):
    @functools.wraps(func)
    def func_out(image, selem=None, *args, **kwargs):
        if selem is None:
            selem = _default_selem(image.ndim)
        return func(image, selem=selem, *args,
                    **kwargs)
    return func_out
```

Beam Search Generation:

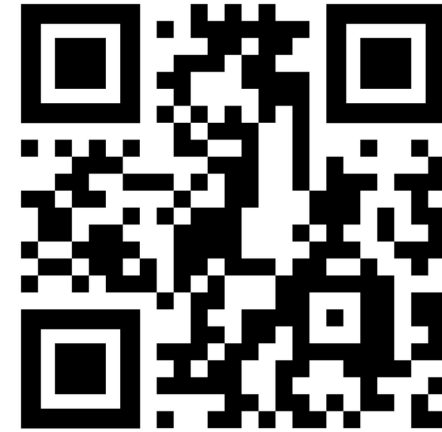
1. decorator to add a default structuring element to morphological operations.
2. decorator to add a default structuring element to morphology functions.
3. decorator to add a default structuring element to binary images.
4. decorator to add a default structuring element to binary image.

Variational Prefix Tuning Generation:

1. decorator to add a default structuring element to morphological operations.
2. generates a binary structure for the given selem.
3. decorator to generate a binary structure of the image if no selem is specified
4. return a decorator that generates a binary structure for the given selem.



Thank You!



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Code